Damandeep Singh

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Summary

Software engineer and MS CS student at Texas A&M with 3 years shipping production backend systems. Work spans distributed systems, LLM agent tooling, and applied machine learning research.

Education

Texas A&M University — College Station, TX, USA
M.S. in Computer Science (Thesis-based) — *Aug. 2026 – May 2028 (Expected)*
- *Coursework:* Machine Learning (CSCE 633), Analysis of Algorithms (CSCE 629), Natural Language Processing (CSCE 689)

National Institute of Technology Srinagar — Srinagar, India
B.Tech. in Information Technology; CGPA: 8.31/10 (Top 10%) — *July 2019 – June 2023*

Experience

Graduate Student Researcher — Aug. 2026 – Present
STMI Lab, Texas A&M University — *College Station, TX, USA*
- Researching multimodal deep learning and large language model methods for clinical risk prediction, fusing EHR time-series with wearable biosignal (ECG/PPG) data into unified patient representations.
- Building retrieval-augmented LLM pipelines and benchmarking model reasoning against clinician-validated outcomes.

Software Development Engineer — July 2024 – June 2026
MediBuddy — *Bengaluru, India*
- Built a natural-language-to-SQL query system on the Model Context Protocol (MCP) with agent-to-agent context management and LLM-driven schema introspection, letting non-technical teams self-serve analytics; shipped behind a Next.js frontend.
- Built a billing reconciliation system using FIFO queue-based idempotency and transactional state machines to eliminate duplicate refunds and pricing errors, reducing monthly financial leakage by 89% (INR 600K to 65K), with Apache Superset dashboards for monitoring.
- Designed a multi-criteria vendor assignment system using a weighted scoring algorithm with a greedy selection heuristic over real-time inventory queries; added Redis caching for sub-100ms latency, cutting medicine delivery time from 48 hours to under 6 hours (60% SLA improvement).
- Developed an automated prescription-validation pipeline using Amazon Textract with transfer learning for valid/invalid classification, cutting manual review time from 10 minutes to real-time.

Software Developer — Sept. 2023 – July 2024
ShipThis — *Bengaluru, India*
- Built a document-parsing pipeline using Azure Document Intelligence with custom OCR models to auto-populate structured data from unstructured emails, and designed GitHub Actions CI/CD pipelines with Jest/Jasmine and ESLint for automated build, test, and deployment.

Research Intern — Dec. 2021 – Mar. 2022
IIT Hyderabad — *Hyderabad, India*
- Built a real-time facial recognition system using MediaPipe FaceMesh for 468-point landmark detection with Euclidean-distance geometric feature extraction, cutting inference latency by 20% through feature-vector and memory optimizations.

Research Intern — Aug. 2021 – Oct. 2021
IIT Delhi — *New Delhi, India*
- Implemented First-Order Model-Agnostic Meta-Learning (MAML) for few-shot N-way K-shot classification with nested gradient-descent loops for meta-optimization, and benchmarked convergence and generalization against the Reptile algorithm.

Projects

Sentinel: MCP Security Gateway | *Go, OpenFGA* | github.com/Damandeep02/sentinel-mcp-gateway
- Built a protocol-aware gateway in Go that intercepts MCP JSON-RPC traffic for tool listing, tool invocation, and resource reads between AI agent clients and tool servers, enforcing per-agent authorization through a pluggable policy engine backed by in-memory rules or an OpenFGA adapter.
- Added fail-closed intent validation for high-risk tool calls and a JSON redactor that strips emails, payment cards, tokens, and secrets from both request parameters and response payloads, with unit tests covering tool filtering, policy denial, intent denial, and redaction.

Credit Card Fraud Detection using Federated Learning | *FedAvg, PyTorch*
- Implemented privacy-preserving fraud detection using the Federated Averaging (FedAvg) algorithm, aggregating local model updates from distributed clients so raw transaction data never left any client and data locality was preserved.
- Evaluated on imbalanced, heterogeneous (non-IID) transaction data distributed across multiple clients, addressing the class-imbalance and client-heterogeneity challenges central to federated settings.

Technical Skills

Languages: Python, TypeScript, JavaScript, Go, Java, C++, SQL
ML / AI: PyTorch, TensorFlow, scikit-learn, LangChain, RAG, MCP, LLMs, OpenCV, NumPy, Pandas
Backend & Infra: Node.js, NestJS, Express.js, Next.js, React.js, PostgreSQL, MongoDB, Redis, Kafka, Docker, AWS (Lambda, SQS, S3), Azure

Honors & Achievements

- **2nd Place, MediBuddy AI/ML Hackathon:** Built Doctor Copilot, an LLM-based clinical decision-support proof of concept.
- **Winner, Devfolio Base Hackathon (Oct. 2024):** Selected as 1 of 16 winning teams across 4 countries for a decentralized gaming platform.